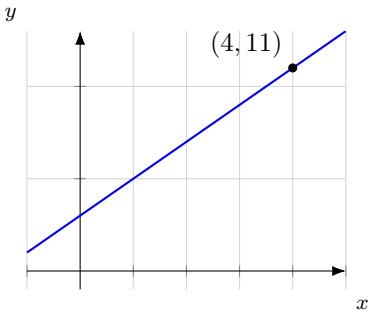
Finding the Equation of a Graph

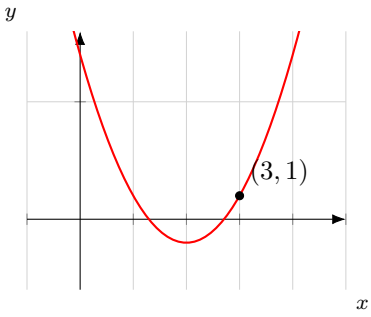
$y = mx + c$ (linear, from the gradient and y-intercept) · $y = a(x-h)^2 + k$ (quadratic, from the turning point (h, k)) · $y = a(x-p)(x-q)$ (quadratic, from the x-intercepts p, q). A y-intercept, x-intercept, or turning point is **not** marked with a dot — read it directly from where the graph crosses or turns on the grid. Only the *extra* point needed to find a (or the gradient) is marked with a dot. Find any unknown constant by substituting that point, then write the full equation. Coordinates are written out for Questions 1–4; for Questions 5–10, the marked point sits exactly on a grid intersection, so read its coordinates from the grid.

1. Find the equation of the line shown.



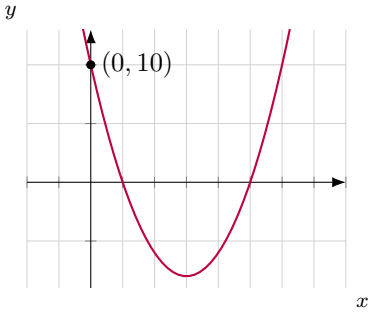
Equation: _____

2. Find the equation of the curve shown.



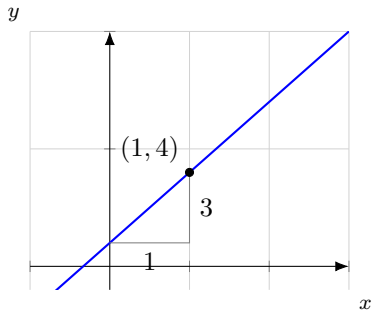
Equation: _____

3. Find the equation of the curve shown.



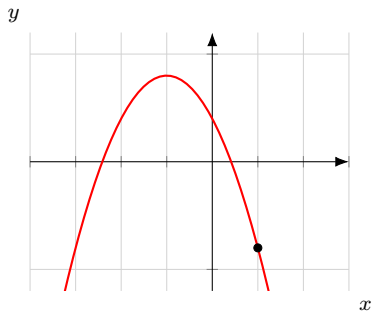
Equation: _____

4. Find the equation of the line shown. The triangle marks the gradient (rise over run).



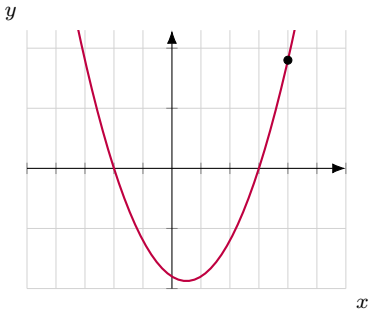
Equation: _____

5. Find the equation of the curve shown.



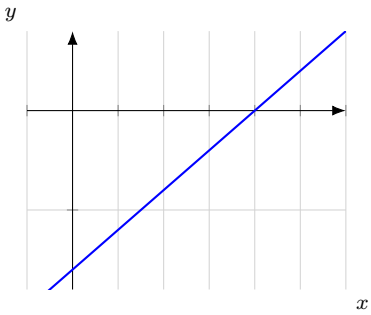
Equation: _____

6. Find the equation of the curve shown.



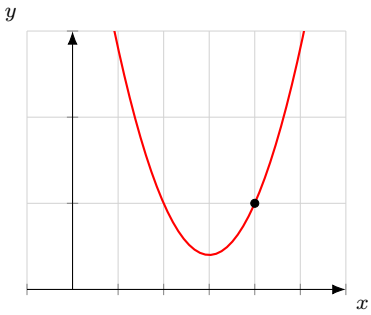
Equation: _____

7. Find the equation of the line shown.



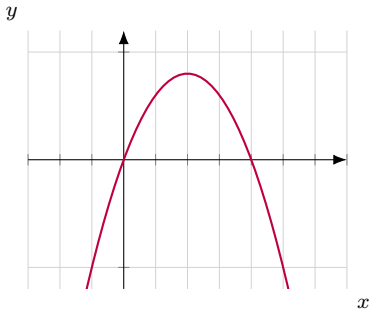
Equation: _____

8. Find the equation of the curve shown.



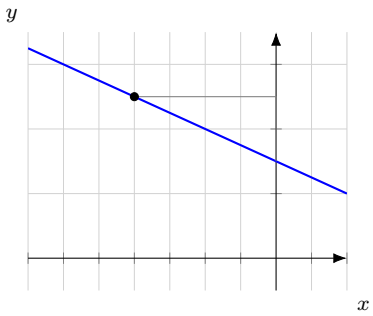
Equation: _____

9. Find the equation of the curve shown.



Equation: _____

10. Find the equation of the line shown. The triangle marks the gradient (rise over run).



Equation: _____